```
import numpy as np  # linear algebra
import pandas as pd  # data preprocessing
import matplotlib.pyplot as plt
import seaborn as sns
```

```
df = pd.read_csv('wine_data.csv', header = None, usecols = [0,1,2])
df.columns = ['Class label', 'Alcohol', 'Malic acid']
```

```
df
```

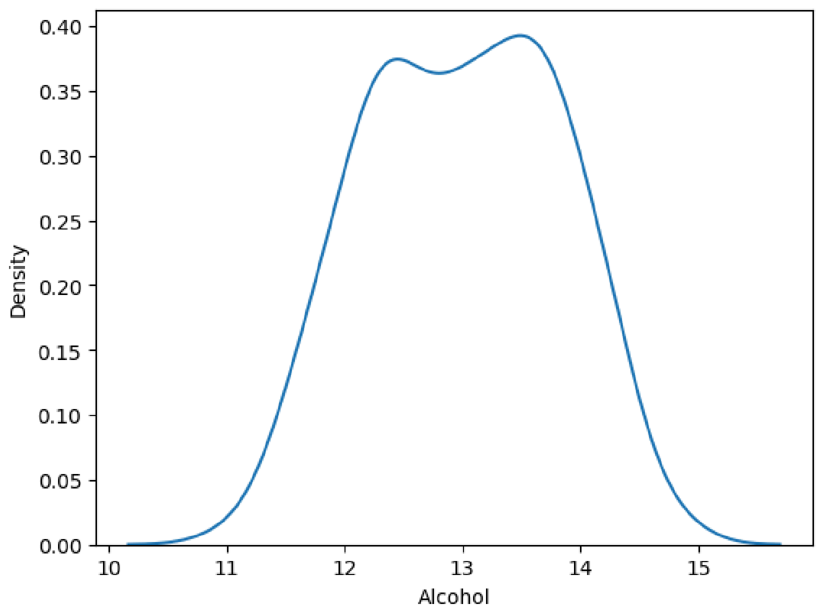
	Class label	Alcohol	Malic acid
0	1	14.23	1.71
1	1	13.20	1.78
2	1	13.16	2.36
3	1	14.37	1.95
4	1	13.24	2.59
...	...	...	...
173	3	13.71	5.65
174	3	13.40	3.91
175	3	13.27	4.28
176	3	13.17	2.59
177	3	14.13	4.10

178 rows × 3 columns

Next steps: Generate code with df | New interactive sheet

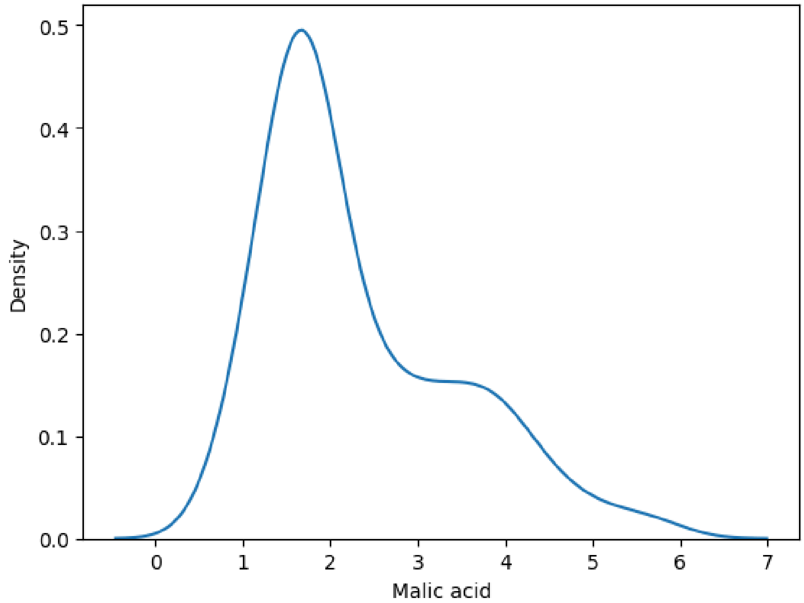
```
sns.kdeplot(df['Alcohol'])
```

```
<Axes: xlabel='Alcohol', ylabel='Density'>
```



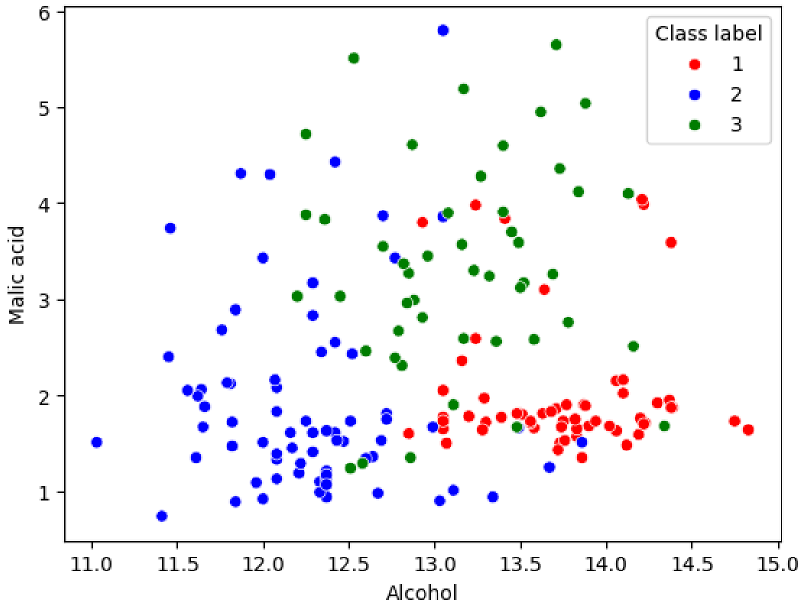

```
sns.kdeplot(df['Malic acid'])
```

```
<Axes: xlabel='Malic acid', ylabel='Density'>
```




```
color_dict = {1:'red', 2:'blue', 3:'green'}
sns.scatterplot(x = 'Alcohol', y = 'Malic acid', data = df, hue = 'Class label', palette = color_dict)
```

```
<Axes: xlabel='Alcohol', ylabel='Malic acid'>
```




```
from sklearn.model_selection import train_test_split
X_train, X_test, y_train, y_test = train_test_split(df.drop('Class label', axis = 1),
                                                    df['Class label'],
                                                    test_size = 0.3,
                                                    random_state = 0)
X_train.shape, X_test.shape
```

```
((124, 2), (54, 2))
```

```
from sklearn.preprocessing import MinMaxScaler
scaler = MinMaxScaler()
# fit the scaler to the train set, it will learn the parameters
scaler.fit(X_train)
# transform train and test sets
X_train_scaled = scaler.transform(X_train)
X_test_scaled = scaler.transform(X_test)
```

```
X_train_scaled = pd.DataFrame(X_train_scaled, columns = X_train.columns)
X_test_scaled = pd.DataFrame(X_test_scaled, columns = X_test.columns)
```

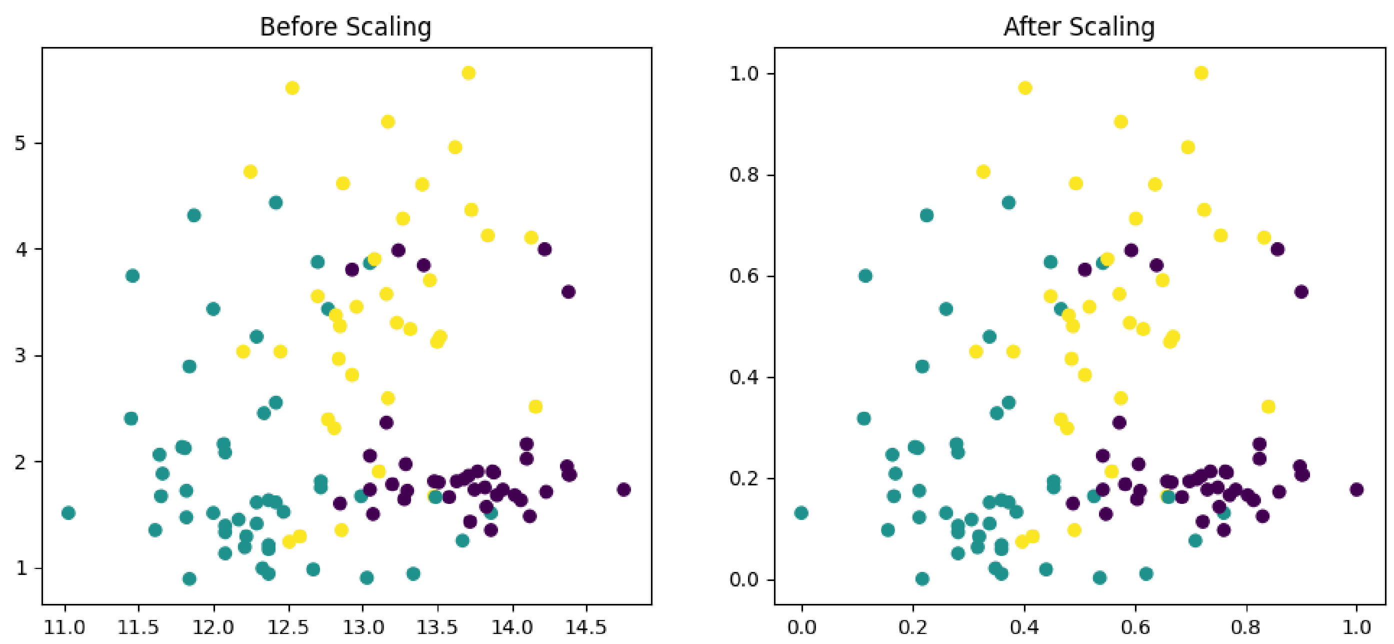
```
np.round(X_train.describe(), 1)
```

	Alcohol	Malic acid
count	124.0	124.0
mean	13.0	2.4
std	0.8	1.1
min	11.0	0.9
25%	12.4	1.6
50%	13.0	1.9
75%	13.6	3.2
max	14.8	5.6

```
np.round(X_train_scaled.describe(), 1)
```

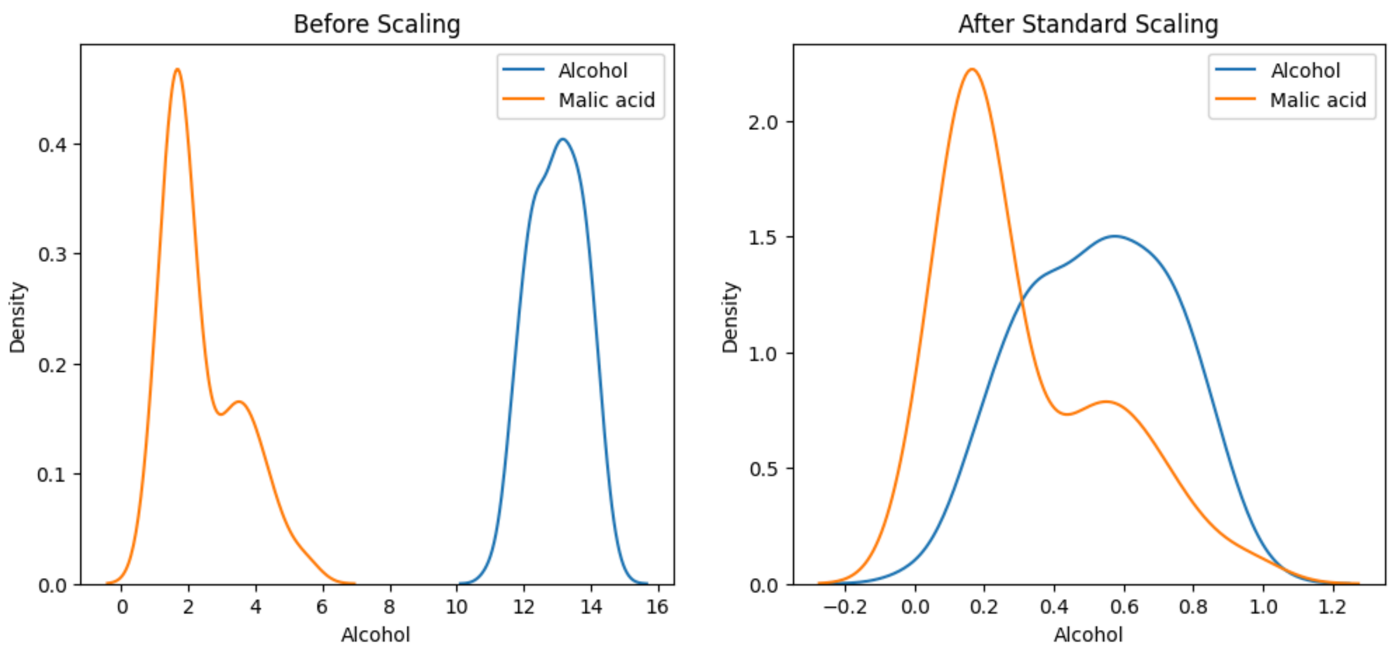
	Alcohol	Malic acid
count	124.0	124.0
mean	0.5	0.3
std	0.2	0.2
min	0.0	0.0
25%	0.4	0.2
50%	0.5	0.2
75%	0.7	0.5
max	1.0	1.0

```
fig, (ax1, ax2) = plt.subplots(ncols = 2, figsize = (12, 5))
ax1.scatter(X_train['Alcohol'], X_train['Malic acid'], c = y_train)
ax1.set_title('Before Scaling')
ax2.scatter(X_train_scaled['Alcohol'], X_train_scaled['Malic acid'], c = y_train)
ax2.set_title('After Scaling')
plt.show()
```

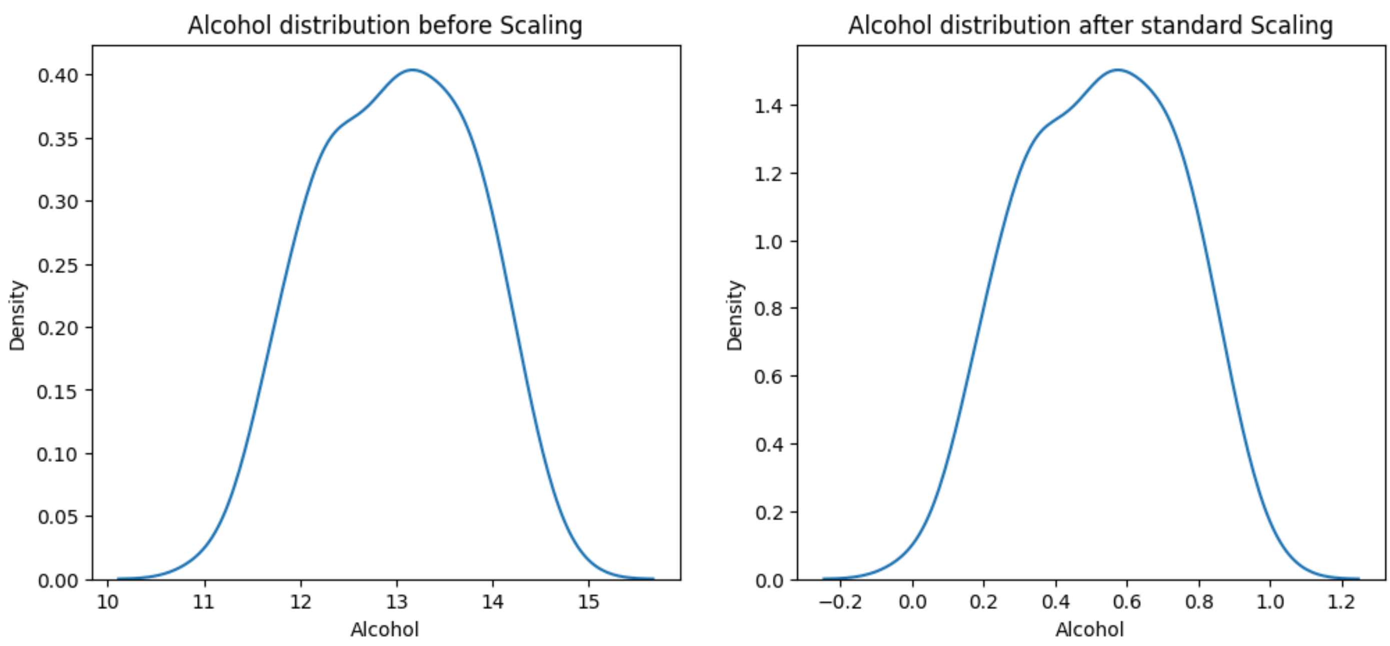



```
fig, (ax1, ax2) = plt.subplots(ncols = 2, figsize = (12, 5))
# before scaling
ax1.set_title('Before Scaling')
sns.kdeplot(X_train['Alcohol'], ax = ax1, label = 'Alcohol')
sns.kdeplot(X_train['Malic acid'], ax = ax1, label = 'Malic acid')
ax1.legend()

# after scaling
ax2.set_title('After Standard Scaling')
sns.kdeplot(X_train_scaled['Alcohol'], ax = ax2, label = 'Alcohol')
sns.kdeplot(X_train_scaled['Malic acid'], ax = ax2, label = 'Malic acid')
ax2.legend()
```

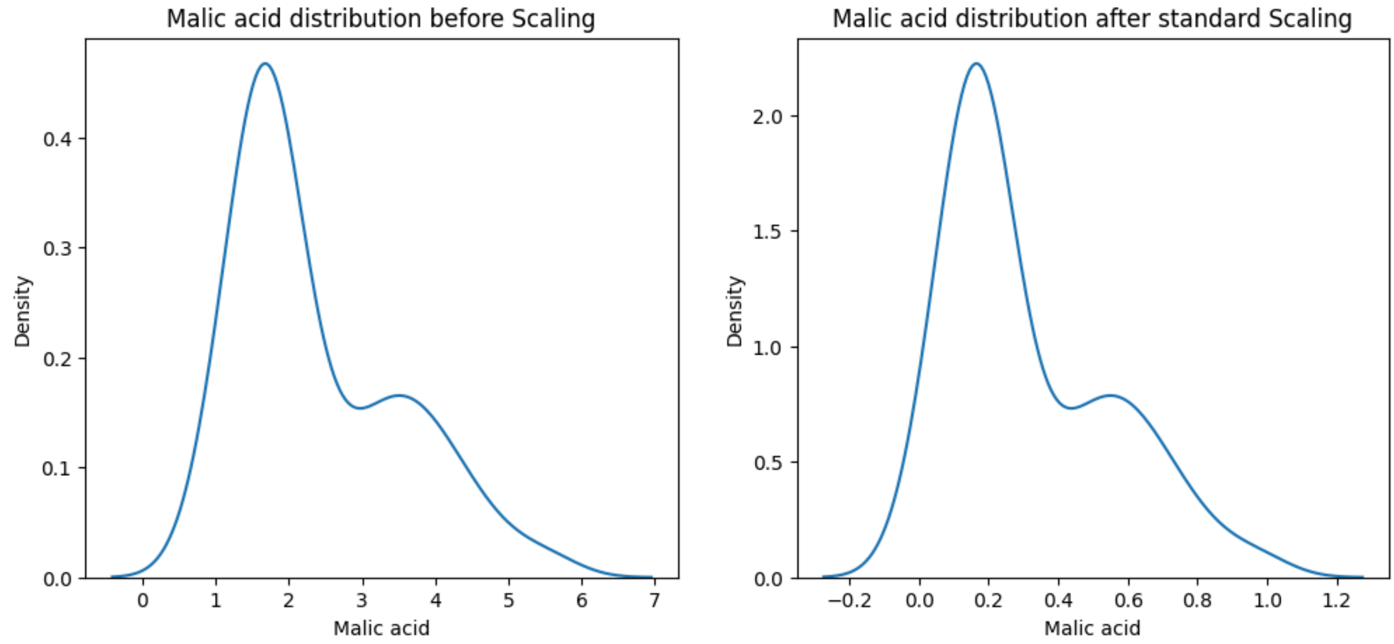
```
plt.show()
```




```
fig, (ax1, ax2) = plt.subplots(ncols = 2, figsize = (12, 5))
# before Scaling
ax1.set_title('Alcohol distribution before Scaling')
sns.kdeplot(X_train['Alcohol'], ax = ax1)

# After Scaling
ax2.set_title('Alcohol distribution after standard Scaling')
sns.kdeplot(X_train_scaled['Alcohol'], ax = ax2)
plt.show()
```




```
fig, (ax1, ax2) = plt.subplots(ncols = 2, figsize = (12, 5))
# before Scaling
ax1.set_title('Malic acid distribution before Scaling')
sns.kdeplot(X_train['Malic acid'], ax = ax1)

# After Scaling
ax2.set_title('Malic acid distribution after standard Scaling')
sns.kdeplot(X_train_scaled['Malic acid'], ax = ax2)
plt.show()
```


Malic acid distribution before Scaling
Density
Malic acid
0.0
0.1
0.2
0.3
0.4
0
1
2
3
4
5
6
7
Malic acid distribution after standard Scaling
Density
Malic acid
0.0
0.5
1.0
1.5
2.0
−0.2
0.0
0.2
0.4
0.6
0.8
1.0
1.2